Bank Loan Analysis

Python Project Report

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Project Overview & Data Loading

Python Libraries Used

This project leverages Python's powerful data science ecosystem for comprehensive loan analysis. The following libraries were imported to handle different aspects of the analysis:

- **pandas (pd)** - For data loading and manipulation
- **numpy (np)** - For advanced numerical aggregation
- **matplotlib.pyplot (plt)** - For building charts
- **seaborn (sns)** - For statistical visualizations
- **plotly.express (px)** - For interactive charts
- **warnings** - For handling correct warnings during errors

Data Source

The dataset was imported from an Excel file named **Financial_loan.xlsx**, loaded directly from the local system into a Jupyter Notebook environment. This file contains comprehensive loan application data that enables detailed financial analysis.

The metadata analysis revealed the structure and characteristics of the dataset, providing insights into data types, missing values, and overall data quality before proceeding with the analysis.

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Key Financial Metrics

100%

Total Loan Applications

Complete dataset coverage showing all loan applications processed

100%

MTD Applications

Month-to-date total loan applications tracking current period performance

100%

Funded Amount

Total amount funded across all loan applications in millions

100%

MTD Funded

Month-to-date total funded amount tracking current funding trends

The analysis tracks both cumulative and month-to-date metrics for loan applications and funded amounts. These key performance indicators provide insights into lending activity patterns and funding trends over time.

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Revenue & Interest Analysis

Total Amount Received

Cumulative revenue collected from all loan applications, measured in millions of dollars

MTD Amount Received

Month-to-date total amount received, tracking current period revenue collection

Average Interest Rate

The average interest rate across all loan applications provides insight into the pricing strategy and revenue generation potential. This metric helps assess the overall profitability of the loan portfolio.

Debt to Income Ratio (DTI)

The DTI ratio measures the relationship between a borrower's total debt and their income. This critical risk metric helps evaluate creditworthiness and default probability, with lower ratios indicating better financial health.

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Loan Quality Metrics

Good Loan Metrics

Good loans represent performing accounts with acceptable risk profiles and expected returns. These metrics include:

- Lower default probability
- Consistent payment history
- Favorable debt-to-income ratios
- Stable employment verification
- Appropriate loan-to-value ratios

Good loans contribute positively to portfolio performance and generate expected interest revenue.

Bad Loan Metrics

Bad loans represent non-performing or high-risk accounts requiring special attention. These metrics include:

- Higher default probability
- Past due payment status
- Elevated debt-to-income ratios
- Unstable employment history
- Insufficient collateral coverage

Bad loans require risk mitigation strategies and may impact overall portfolio performance.

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Monthly Trends by Issue Date

The time series analysis tracks three critical metrics over time, revealing seasonal patterns and growth trends in the loan portfolio.

Total Amount Funded

Monthly trend showing funding volume patterns and seasonal variations

Total Amount Received

Monthly trend tracking revenue collection and payment patterns

Total Loan Applications

Monthly trend indicating application volume and customer demand

These time series visualizations help identify peak periods, growth trends, and seasonal patterns in lending activity. The analysis syntax provided enables recreation of these charts in Jupyter Notebook environments.

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Regional Analysis by State

Geographic Distribution of Funded Amount

The regional analysis breaks down total funded amounts by state, revealing geographic patterns in lending activity. This visualization helps identify high-volume regions and potential market opportunities.

Key insights from regional analysis include:

- Identification of top-performing states by funding volume
- Geographic concentration patterns in loan distribution
- Regional risk exposure assessment
- Market penetration analysis across different areas
- Opportunities for geographic expansion or focus

The column chart visualization syntax provided enables detailed exploration of state-by-state funding patterns.

Regional Insights

Understanding geographic distribution helps optimize resource allocation and identify regional risk factors.

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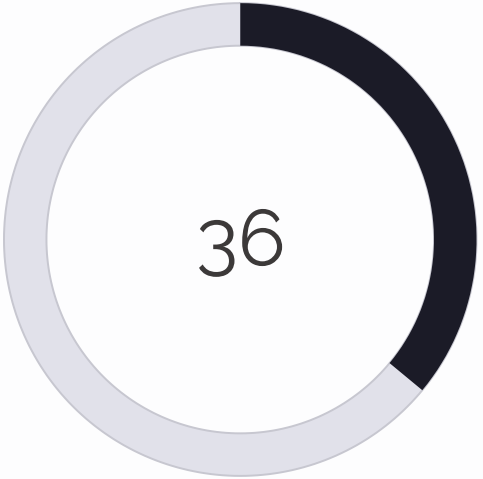
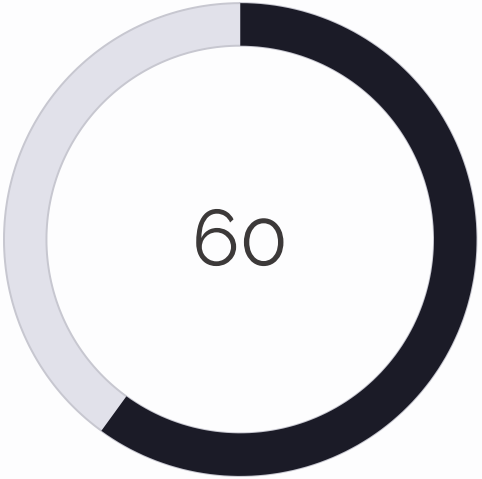
Loan Term Analysis



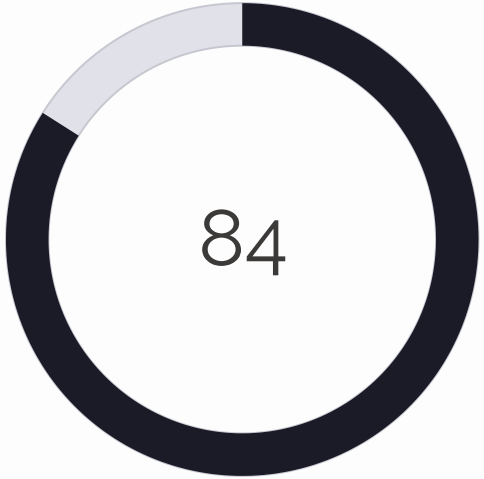
Months

Short-term loan products



Months

Medium-term loan products



Months

Long-term loan products

The loan term analysis examines total funded amounts distributed across different loan durations. This donut chart visualization reveals the product mix and customer preferences for various loan terms.

Understanding term distribution helps assess interest rate risk, liquidity requirements, and customer behavior patterns. Shorter terms typically have lower risk but also lower interest income, while longer terms provide higher revenue potential with increased default risk exposure.

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Employment & Loan Purpose Analysis

Employment Length Analysis

Total funded amount distribution by employment tenure, showing how job stability correlates with loan amounts. The column bar chart visualization syntax enables detailed exploration of employment history patterns.

Loan Purpose Analysis

Total funded amount by stated loan purpose, revealing customer motivations and usage patterns. The bar chart visualization syntax enables detailed exploration of purpose-based funding distribution.

These analyses provide insights into borrower characteristics and loan usage patterns. Employment length serves as a proxy for income stability and repayment capacity, while loan purpose reveals customer needs and potential risk factors associated with different use cases.

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Home Ownership Analysis

Home Ownership by Total Funded Amount

The home ownership analysis examines funded amounts distributed across different property ownership statuses. This treemap chart visualization reveals the relationship between property ownership and loan amounts.

Key ownership categories typically analyzed include:

- **Owned** - Properties owned outright without mortgage
- **Mortgage** - Properties with active mortgage loans
- **Rent** - Renting without property ownership
- **Other** - Alternative ownership arrangements

Property ownership status serves as an important risk factor, with owned properties typically indicating greater financial stability and providing potential collateral for secured loans.

Key Insights

Home ownership patterns reveal important risk and demographic insights for loan portfolio management.

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